

Ralu Chineme (he/him)

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EDUCATION

University of British Columbia

Expected Graduation: May 2029

Bachelor of Applied Science – Computer Engineering

- Co-op Availability: Available for 16 months beginning May 2027
- Relevant Courses: Introduction to Computation in Engineering Design: 94%

TECHNICAL SKILLS

Languages & Programming: C, C++, Java, Python, Typescript, HTML/CSS

Hardware: Arduino, Microcontrollers, PC Building & Assembly

Tools & Platforms: GitHub, VS Code, Linux, Docker, Tailscale, Replit, IntelliJ IDEA

Certifications: Introduction to Microprocessors - Arm Education (via edx.org), Standard First Aid + CPR-C, HCS 3000

TECHNICAL WORK EXPERIENCE

MFC Games | Calgary, AB

May 2024 – August 2024

Website Administrator

- Regularly maintained and updated the organization's website with blog posts, event information, and photo galleries, ensuring visitors had accurate and up-to-date information throughout the summer
- Collaborated with organizers to coordinate website updates, improving site navigation and user experience
- Communicated effectively with organizers on content and design priorities, keeping messages consistent across the site

TECHNICAL PROJECTS

Raspberry Pi Homelab | Personal Project

April 2026 - Present

- Engineered a multi-service Raspberry Pi 5 homelab, deploying six containerized applications (Nextcloud, Pi-hole, Jellyfin, Uptime Kuma, Portainer, Homepage) to run reliably on a single device
- Configured an NVMe SSD with dedicated storage paths for cloud and media, compartmentalizing application data from the OS drive to protect against data loss
- Implemented Pi-hole as a home network DNS server and Tailscale for remote access, extending Nextcloud into a multi-user private cloud so family members could access files from outside the home network
- Administered the headless Linux server through SSH, troubleshooting Docker, networking, storage and authentication issues to keep services online and accessible

Calorie Tracker App | Personal Project

December 2024 – October 2025

- Built a web-based calorie tracking application using Replit and AI-assisted development tools, delivering a functioning nutrition tracker
- Evaluated and modified AI-generated TypeScript code, catching and fixing implementation concerns before deployment
- Refined CSS and front-end design to enhance usability, resulting in a more polished UI

Arduino Servo Claw | APSC 101, University of British Columbia

2026

- Collaborated with a group of students to design and build a sensor-driven mechanical claw, contributing the rotation-control code and physical construction
- Tuned the servo's rotation range and the sensor's trigger distance to grip effectively, avoiding over-rotation and inconsistent detection

OTHER WORK EXPERIENCE

Master's Academy and College | Calgary, AB

January 2022 – June 2024

Junior High Math Tutor

- Provided one-on-one tutoring in Grade 8 mathematics to students, adapting explanations to learning styles to help students digest information
- Guided students through practice problems, helping them build problem-solving skills and confidence

Pedalheads Bike Camps | Calgary, AB

Summer 2022, 2023, 2026

Camp Instructor

- Taught cycling skills and safety drills to groups of children, building their confidence and cycling ability each camp day
- Adapted lessons to a range of skill levels, encouraging participation and strengthening learning
- Communicated effectively with parents to ensure a positive experience and environment

VOLUNTEER EXPERIENCE

Calgary Public Library | Calgary, AB

December 2023 – June 2024

Coding Buddies & Reading Buddies Volunteer

- Mentored young children in basic block coding concepts through the library's Coding Buddies program, adapting explanations to different skill levels to build participants understanding of programming fundamentals
- Supported early literacy development as a Reading Buddies volunteer, working with children on reading skills

Master's Eagles Junior High Basketball | Calgary, AB

December 2023 – June 2025

Coach

- Co-coached the Grade 7 Boys' basketball team to a CISAA league championship, running 3-4 practices a week and 1-2 games weekly
- Communicated effectively with players and parents while promoting discipline and leadership

AWARDS

Athlete of the Year | Master's Academy and College

2025

INTERESTS & ACTIVITIES

- Competing in Java/Python programming competitions and hackathons such as MIT Battlecode, Battlesnake, and the JunioTech Hackathon
- Daily weightlifting with a current focus on powerlifting
- Built a custom video editing PC, troubleshooting hardware issues during initial setup
- Exploring microprocessors and semiconductors through online courses and hands-on learning